

Linear Algebra – MAT 2610

Section 4.2 (Null Spaces, Column Space and Linear Transformations)

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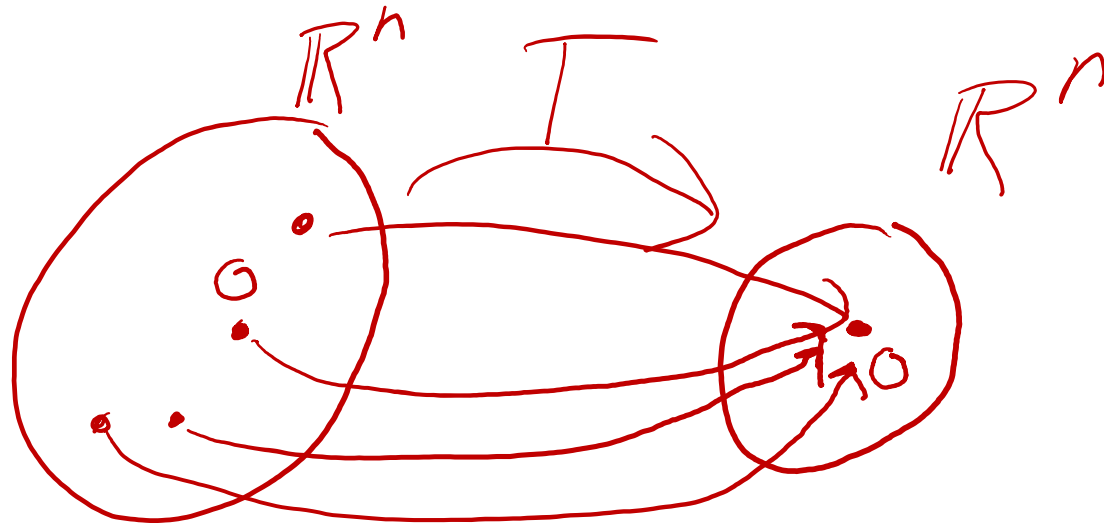


Recall: The **null space** of an $m \times n$ matrix A is the set of solutions to $Ax = 0$, denoted $\text{Null}(A)$ or $\text{Nul}(A)$

$$\text{Null}(A) = \{x: x \in \mathbb{R}^n \text{ and } Ax = 0\}$$

When considered as a transformation, the **null space** of a matrix A is the set of all $x \in \mathbb{R}^n$ that are mapped into the zero vector of $\mathbb{R}^m$ via the linear transformation $x \mapsto Ax$

$$T: \mathbb{R}^n \rightarrow \mathbb{R}^m$$



Theorem 2

The null space of an $m \times n$ matrix A is a subspace of $\mathbb{R}^n$. Equivalently, the set of all solutions to a system $Ax = 0$ of m homogeneous linear equations in n unknowns is a subspace of $\mathbb{R}^n$.

$$\begin{array}{ccc} \begin{array}{c} m \text{ rows} \\ \left[\begin{array}{c} \\ \\ \\ \\ \end{array} \right] \end{array} & \begin{array}{c} n \text{ cols} \\ \left[\begin{array}{c} \\ \\ \\ \\ \end{array} \right] \end{array} & \begin{array}{c} n \times 1 \in \mathbb{R}^n \\ \left[\begin{array}{c} \\ \\ \\ \\ \end{array} \right] \end{array} \end{array} \rightarrow \begin{array}{c} m \times 1 \in \mathbb{R}^m \\ \left[\begin{array}{c} 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{array} \right] \end{array}$$



Proof of Theorem 2

1. The zero vector is in $\text{Null}(A)$
2. Whenever u and v are in $\text{Null}(A)$, then so is $u + v$
3. Whenever u is in $\text{Null}(A)$ and c is a scalar, then cu is also in $\text{Null}(A)$

1) Is $0 \in \text{Null}(A)$, or is $A0 = 0$ ✓

2) $u, v \in \text{Null}(A)$ means $Au = 0$ and $Av = 0$

what about $A(u+v)$? $A(u+v) = Au + Av = 0 + 0 = 0$ ✓

3) $u \in \text{Null}(A)$ means $Au = 0$

what about $A(cu)$? $A(cu) = c(Au) = c0 = 0$ ✓

All 3 properties hold, so $\text{Null}(A)$ is a subspace



Example: Find an explicit vector description of the null space of $A = \begin{bmatrix} 1 & 2 & 1 & 3 & 8 \\ 2 & 4 & 1 & 0 & 8 \\ 3 & 6 & 2 & 3 & 16 \end{bmatrix}$

Note that A row reduces to

$$\begin{array}{ccccc} & \downarrow & & \downarrow & \\ x_1 & x_2 & x_3 & x_4 & x_5 \\ \begin{bmatrix} 1 & 2 & 1 & 0 & 4 \\ 0 & 0 & 0 & 1 & 4/3 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} \end{array}$$

$$\left. \begin{array}{l} x_1 + 2x_2 + x_3 + 4x_5 = 0 \\ x_4 + 4/3 x_5 = 0 \end{array} \right\} \begin{array}{l} x_1 = -2x_2 - x_3 - 4x_5 \\ x_2 = x_2 \\ x_3 = x_3 \\ x_4 = -4/3 x_5 \\ x_5 = x_5 \end{array}$$

$$\begin{aligned} \text{Null}(A) = & x_2 \begin{bmatrix} -2 \\ 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} + x_3 \begin{bmatrix} -1 \\ 0 \\ 1 \\ 0 \\ 0 \end{bmatrix} \\ & + x_5 \begin{bmatrix} -4 \\ 0 \\ 0 \\ -4/3 \\ 1 \end{bmatrix} \end{aligned}$$



Definition: The column space of an $m \times n$ matrix A is the span of its columns, denoted $col(A)$. If $A = [a_1, \dots, a_n]$ then

$$Col(A) = Span\{a_1, \dots, a_n\}$$

$$Col(A) = \{b : Ax = b \text{ for some } x \in \mathbb{R}^n\}$$

$b \in \mathbb{R}^m$

$$c_1 a_1 + c_2 a_2 + \dots + c_n a_n = b$$



Theorem 3

The column space of an $m \times n$ matrix A is a subspace of ~~$\mathbb{R}^n$~~ $\mathbb{R}^m$

$$\text{col}(A) = \text{Span} \{a_1, a_2, \dots, a_n\} \text{ where}$$

a_i are columns of A



Note

If A is an $m \times n$ matrix A then:

- a. The column space of A is a subspace of R^m
- b. The null space of A is a subspace of R^n



Linear Transformation

Definition: If V and W are vector spaces, then a **linear transformation** $T: V \rightarrow W$ is a function where the following holds for any vectors $u, v \in R^n$ and c any scalar:

1. $T(u + v) = T(u) + T(v)$
2. $T(cu) = cT(u)$



The **kernel** of a linear transformation $T: V \rightarrow W$ is the set of vectors $u \in V$ such that $T(u) = 0$

→ similar to Null space

The **range** of a linear transformation $T: V \rightarrow W$ is the set of all vectors $w \in W$ where $T(u) = w$ for some $u \in V$

→ similar to Col space

$$T: P^4 \rightarrow \mathbb{R}^5$$

where $T(a_1 + a_2x + a_3x^2 + a_4x^3 + a_5x^4) =$

$$\begin{bmatrix} a_1 \\ a_2 \\ a_3 \\ a_4 \\ a_5 \end{bmatrix}$$



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